



Lecture 1: Probability and axioms

Yuan Shen

shenyuan_ee@tsinghua.edu.cn

Tsinghua University

February 23, 2021

- Staff:
 - Lecturer: Yuan Shen, shenyuan_ee@tsinghua.edu.cn
 - TAs:
 - Ranhao Zhang, zrh20@mails.tsinghua.edu.cn
 - Yuchao Chen, cyc20@mails.tsinghua.edu.cn
 - Mingtao Huang, huangmt19@mails.tsinghua.edu.cn
- Lecture:
 - Tuesday, 9:50 - 11:25 am, Building 5201
 - Online: Tencent Meeting, 463 3415 3181
- Recitation Session:
 - Wed./Thu./Fri., 19:30 - 20:30 pm, Rohm 8-208, every week
 - Review important concepts and exercise problems
 - Three sessions cover the same materials
- Textbook: Intro. to Probability, 2nd ed. (Bertsekas/Tsitsiklis)

- Office hour:
 - Friday 3 - 4 pm, 11-201, Rohm building
 - You can also make an appointment via Email or WeChat
- Grading:
 - Weekly homework (30%):
 - Submit online via Web Learning
 - 50%-off for late submission
 - Mid-term exam (25%)
 - Week 9, in-class
 - Coverage: Lecture 1 to 7
 - Final exam (40%)
 - Participation (5% + bonus 5%)

Our course uses materials of MIT's Probabilistic Systems Analysis and Applied Probability for reference.

- **Global Hybrid Classroom**

- Open course for students from St Petersburg University, Nanyang Technological University, and Politecnico di Milano
- Attend online via Tencent Meeting
- Audit or Credit
- Promoting international teaching cooperation & Innovating new types of learning



1. Probabilistic models
2. Sample space and events
3. Axioms of probability
4. Recitation
5. Summary



1. Probabilistic models
2. Sample space and events
3. Axioms of probability
4. Recitation
5. Summary



- A mathematical description of an uncertain situation

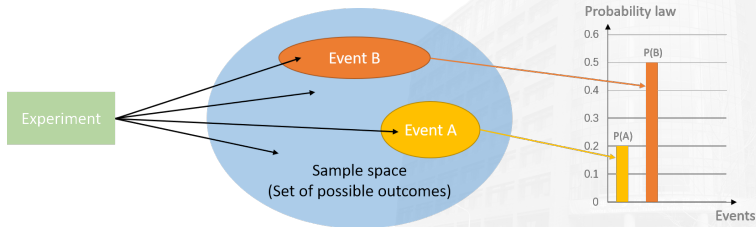
Elements of a Probabilistic Model

- ① **Sample space Ω :** the set of all possible *outcomes* of an experiment
- ② **Probability law $\mathbf{P}$:** assigns to a set A of possible outcomes (called *event*) a nonnegative number $\mathbf{P}(A)$ (called *probability*) that encodes our knowledge of belief about the collective “likelihood” of the elements of A .

- A mathematical description of an uncertain situation

Elements of a Probabilistic Model

- ❶ **Sample space Ω :** the set of all possible *outcomes* of an experiment
- ❷ **Probability law $\mathbf{P}$:** assigns to a set A of possible outcomes (called *event*) a nonnegative number $\mathbf{P}(A)$ (called *probability*) that encodes our knowledge of belief about the collective “likelihood” of the elements of A .



1. Probabilistic models
2. Sample space and events
3. Axioms of probability
4. Recitation
5. Summary

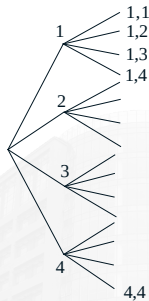
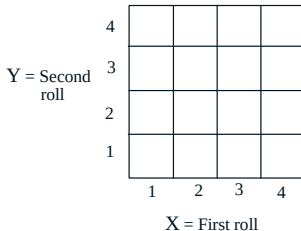


- Every probabilistic model involves an underlying process, called the **experiment**, that will produce **exactly one** out of several possible **outcomes**
- **Sample space** Ω : the set of all possible outcomes
- **Event**: a subset of the sample space, i.e., a collection of possible outcomes

- Every probabilistic model involves an underlying process, called the **experiment**, that will produce **exactly one** out of several possible **outcomes**
- **Sample space** Ω : the set of all possible outcomes
- **Event**: a subset of the sample space, i.e., a collection of possible outcomes
- Sample space Ω must be:
 - **Mutually exclusive**: unique outcome
 - **Collectively exhaustive**: outcome included
- **Art**: At the “right” granularity – enough detail to distinguish between all outcomes *of interest*, while avoiding irrelevant details.

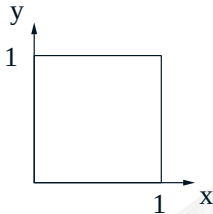
Sample space: Discrete example

- Two (sequential) rolls of a tetrahedral die
 - Sample space vs. sequential description



Sample space: Continuous example

- $\Omega = \{(x, y) \mid 0 \leq x, y \leq 1\}$



1. Probabilistic models
2. Sample space and events
3. Axioms of probability
4. Recitation
5. Summary



- Probability $\mathbf{P}(A)$ is assigned to event A with certain properties, which intuitively specifies the “likelihood” of any outcome, or of any set of possible outcomes

Probability Axioms

- ① **Nonnegativity:** $\mathbf{P}(A) \geq 0$
- ② **Normalization:** $\mathbf{P}(\Omega) = 1$
- ③ **Additivity:** If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

- Probability $\mathbf{P}(A)$ is assigned to event A with certain properties, which intuitively specifies the “likelihood” of any outcome, or of any set of possible outcomes

Probability Axioms

- ① **Nonnegativity:** $\mathbf{P}(A) \geq 0$
- ② **Normalization:** $\mathbf{P}(\Omega) = 1$
- ③ **Additivity:** If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

Remarks:

- Axiom 3 can be strengthened: **countable** number of elements
- Do weird sets have probabilities? **

Discrete probability law

If Ω consists of a **finite** number of possible outcomes, then the probability of any event $\{s_1, s_2, \dots, s_n\}$ is the sum of the probabilities of its elements

$$\mathbf{P}(\{s_1, s_2, \dots, s_n\}) = \mathbf{P}(s_1) + \dots + \mathbf{P}(s_n)$$

- **Discrete uniform law:** If all sample points be equally likely, then

$$\mathbf{P}(A) = \frac{\text{\# of elements of } A}{\text{total \# of sample points}}$$

- Computing probabilities $\equiv$ counting
- Examples: Defines fair coins, fair dice, well-shuffled card decks

Example

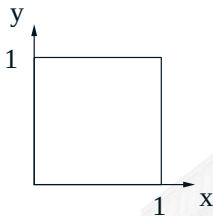
- Let every possible outcome have probability $1/16$
 - $\mathbf{P}((X, Y) \text{ is } (1, 1) \text{ or } (1, 2)) =$
 - $\mathbf{P}(\{X = 1\}) =$
 - $\mathbf{P}(X + Y \text{ is odd}) =$
 - $\mathbf{P}(\min(X, Y) = 2) =$

Y = Second roll

4				
3				
2				
1				
	1	2	3	4

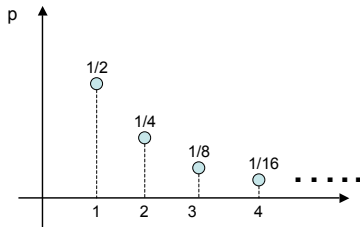
X = First roll

- Compared to discrete models: Single-element events may not be sufficient to characterize the probability law
- Two “random” numbers in $[0, 1]$

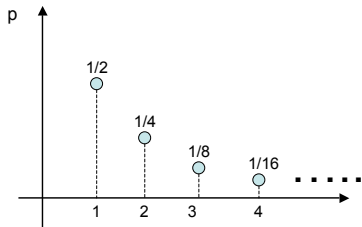


- **Uniform law:** Probability = Area
 - $\mathbf{P}(X + Y \leq 1/2) =$
 - $\mathbf{P}((X, Y) = (0.5, 0.3)) =$

- Sample space: $\{1, 2, \dots\}$
 - We are given $\mathbf{P}(n) = 2^{-n}$, $n = 1, 2, \dots$
 - Find $\mathbf{P}(\text{outcome is even})$



- Sample space: $\{1, 2, \dots\}$
 - We are given $\mathbf{P}(n) = 2^{-n}$, $n = 1, 2, \dots$
 - Find $\mathbf{P}(\text{outcome is even})$



Axiom 3, strengthened version (Countable additivity)

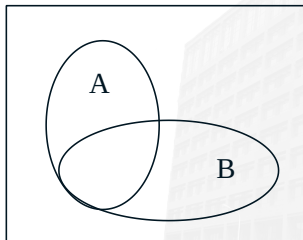
If $A_1, A_2, \dots$ are *disjoint* events, then

$$\mathbf{P}(A_1 \cup A_2 \cup \dots) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \dots$$

Properties:

Consider a probability law, and let A , B and C be events.

- If $A \subset B$, then $\mathbf{P}(A) \leq \mathbf{P}(B)$
- $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B)$
- $\mathbf{P}(A \cup B) \leq \mathbf{P}(A) + \mathbf{P}(B)$
- $\mathbf{P}(A \cup B \cup C) = \mathbf{P}(A) + \mathbf{P}(A^c \cap B) + \mathbf{P}(A^c \cap B^c \cap C)$



- A narrow view: a branch of math
 - Axioms $\implies$ theorems

Theorem: “Frequency” of event A “is” $\mathbf{P}(A)$

- Are probabilities frequencies?
 - $\mathbf{P}(\text{coin toss yields heads}) = 1/2$
 - $\mathbf{P}(\text{Trump wins 2024 US Election}) = 0.1$
 - $\mathbf{P}(\text{a piece of equipment aboard the space shuttle fails}) = 10^{-8}$
- Probability models are a framework for describing uncertainty
 - Use for consistent reasoning
 - Use for predictions, decisions
- **Model vs Reality: tradeoff between accuracy and tractability**

1. Probabilistic models
2. Sample space and events
3. Axioms of probability
- 4. Recitation**
5. Summary



1. Give a mathematical derivation of the formula

$$\mathbf{P}((A \cap B^c) \cup (A^c \cap B)) = \mathbf{P}(A) + \mathbf{P}(B) - 2\mathbf{P}(A \cap B)$$

Your derivation should be a sequence of steps, with each step justified by appealing to one of the probability axioms.

2. Out of the students in a class, 60% are geniuses, 70% love chocolate, and 40% fall into both categories. Determine the probability that a randomly selected student is neither a genius nor a chocolate lover.

3. A six-sided die is loaded in a way that each even face is twice as likely as each odd face. Construct a probabilistic model for a single roll of this die, and find the probability that a 1, 2, or 3 will come up.
4. Romeo and Juliet have a date at a given time, and each will arrive at the meeting place with a delay between 0 and 1 hour, with all pairs of delays being equally likely. The first to arrive will wait for 15 minutes and will leave if the other has not yet arrived. What is the probability that they will meet?

1. Probabilistic models
2. Sample space and events
3. Axioms of probability
4. Recitation
- 5. Summary**



- Sample space
 - List of possible outcomes
- Axioms of probability
 - $\mathbf{P}(A) \geq 0$
 - $\mathbf{P}(\text{universe}) = 1$
 - If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$
- Readings: Section 1.1, 1.2
- Assignment: Problem Set 1
- Next lecture: Conditioning and Bayes' rule